

Sashwat Prasadh

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🌐 LinkedIn

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Education

Memorial University of Newfoundland
Bachelor's of Science in Physics

2019–2023

Experience

Data Scientist: Machine Learning

Memorial University, Department of Biology

May 2024 – Present

- Neural Network Applications for Physical and Biological Oceanography.

Research Assistant- Computational Fluid Dynamics

Memorial University, Department of Physical Oceanography

May 2023 – August 2023

- Data Analysis and Python Simulation of Particle Image Velocimetry.

Experimental Physics Assistant

Memorial University, Department of Chemistry

Sept 2023 – Dec 2023

- Spectroscopy, Lab Safety, and Website Design for a Material Physics Lab.

Research Intern- Astrophysics

Memorial University, Department of Physics

Jan 2024 – April 2024

- Inverse and Forward Modeling using MESA to study binary Cepheids.

Physics Resource Development officer

Memorial University, Department of Physics

Jan 2023 – April 2023

- Workshop Manuals for Facilitating Science Education in School Curriculum.

Publications (Upcoming)

Prasadh, Sashwat; Neilson, H.R.; Evans, N.R. *Resolving Age Discrepancy: Stellar Merger and Evolution of Three Galactic Cepheid Binaries* | Submission: April 2025 | [\[📄\]](#)

Prasadh, Sashwat; Ralph, Rebecca; Fitzgerald, J G.; Wheeler, J.D. *Neural Network Applications for Particle Image Velocimetry of Steady Fluid Flow* | Expected Submission: May 2025 | [\[📄\]](#)

Achievements

- **Dr. Hugh Anderson Junior and Senior Scholarship in Physics and Physical Oceanography** | Awarded 2023
- **Best Poster in Physics & Physical Oceanography** | SEA Conference | April 2024

Conference Presentations

- **Prasadh, Sashwat;** Neilson, H.; Evans, N.R. *Stellar Astrophysics: Age Discrepancy in Cepheid Binaries*, AUPAC, Feb. 2024, Mount Allison University, NB
- Ralph, Rebecca; **Prasadh, S;** Fitzgerald, JG; Wheeler, JD. *Development of machine learning methods to estimate local fluid flow environments surrounding marine pelagic organisms*, Science Atlantic Conference (Applied Aquatic Science), Coauthored Talk - March 2024, Dalhousie University, NS
- **Prasadh, Sashwat;** Neilson, H.; Evans, N.R. *Resolving Age Discrepancy: Stellar Mergers and Evolution of Three Cepheid Binaries*, SEA, April 2024, Memorial University, NL
- Ralph, Rebecca; **Prasadh, S;** Fitzgerald, JG; Wheeler, JD. *Developing new methods for improving estimates of fluid flow velocities in in situ plankton-flow studies*, SEA, April 2024, Memorial University, NL

Technical Skills

- **Programming:** Python, LaTeX, Matlab, Git, Inverse Modelling, Image Analysis, Monte Carlo Simulations
- **Machine Learning:** TensorFlow, PyTorch, Scikit-Learn, Neural Network, Time-Series Forecasting, Statistical Analysis
- **Skills:** Spectroscopy (NMR, FTIR, Brillouin), Data Collection & Visualization, Financial Modeling, Risk Analysis